Track 2 — Drug Repositioning Report

Rare Disease, Real Kid: MVA Hackathon 2026

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Submission date: 24 October 2026
Report type: Literature-based, genotype-stratified repositioning hypotheses (no patient data used — see Section 1)

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1. Transparency statement (please read first)

We want judges to know exactly what this report is and what it is not.

- **Time spent:** about **1 h 6 min** in total, on the deadline day (24 Oct 2026, 22:44–23:50 UTC). A first draft was produced in the first few minutes; the rest of the time went into checking every claim against its source, adding new evidence and rebuilding the document. A short earlier session the same day was spent only reading the hackathon rules. This is a rapid, good-faith contribution, not a months-long analysis.
- **No patient data was used.** We did **not** download or analyse the child's VCF/BAM/CRAM or clinical files. We joined too late for dataset access to be approved, and we also chose not to process patient data with the AI tools we had, because the organizers' guidance (HF Discussion #2) requires Business/API-tier terms for patient-specific data and we were using a personal (consumer) account.
- **AI disclosure:** Anthropic Claude (model Claude Opus 5.5) on a personal consumer account, first in the claude.ai app and then in a Claude Code session, for literature search, synthesis, drafting and document building. In the Claude Code session two parallel Claude sub-agents were used: one searched for additional candidates (Section 3, "fisetin", "avoid HSP90 inhibitors" and "other candidates considered"), and a second was launched to fact-check the draft independently, but it had not finished before the deadline and its output was not used. Reference details were instead cross-checked by the author in a separate Claude Code session on his own laptop. OpenAI Codex (ChatGPT Plus account) was also used, only as an independent checker of reference details; it received no patient data. It received **only public information** (published papers, MedlinePlus, FDA/press announcements). The human author directed the work, set the requirements and approved the content.
- **Consequence:** because we do not know which gene or variant the child carries, the report is **stratified by MVA genotype** (BUB1B, TRIP13, CEP57) and by whether the child's BubR1 protein is unstable, truncated or functionally defective. Anyone with the confirmed variant — for example from a Track 1 submission — can pick the relevant branch from Section 4 immediately.
- **Not medical advice.** Everything here is a hypothesis for expert review and pre-clinical testing. No candidate should be given to the child without his clinical team.
- All references were located and checked during this session; their key claims are summarised in our own words. Each candidate carries an explicit evidence level.

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2. Disease mechanism in brief

Mosaic variegated aneuploidy (MVA) is an autosomal recessive disorder in which a variable fraction of cells carries gains or losses of random whole chromosomes. The established causal genes all act on accurate chromosome segregation in mitosis [1–3]:

Subtype	Gene	Protein role	Key clinical risk
MVA1	**BUB1B**	BubR1, core spindle-assembly-checkpoint (SAC) protein	Embryonal tumours, incl. rhabdomyosarcoma and Wilms tumour [1,4]
MVA3	**TRIP13**	AAA+ ATPase required for SAC function	Wilms tumour in all six first-reported patients [4]
MVA2	**CEP57**	Centrosome/microtubule-associated protein	Not linked to embryonal tumours; only minimal SAC defect in patient cells [4]

Other genes (e.g. BUB1, BUB3) have also been reported [5]. Shared features include pre- and post-natal growth restriction and microcephaly; for BUB1B and TRIP13 the childhood cancer risk is strongly elevated, and patient cells show severe SAC impairment [4].

Why BUB1B disease is pharmacologically interesting. BUB1B variants can reduce transcript abundance, destabilise BubR1 protein, or impair its function without destabilising it. Reference [6] demonstrated unstable kinase-domain missense proteins, but also stable missense proteins with functional defects; transcript loss was established or inferred for particular truncating alleles, not all truncations. Wild-type BUBR1 re-expression restored checkpoint activity in patient cells [6]. In

other words, for some MVA1 patients the problem is partly *not enough stable BubR1*, which is a druggable quantity.

Three findings from mouse models connect this to existing compounds:

1. **Low BubR1 dose drives a progeroid, senescence-rich phenotype.** Mice with ~10% of normal BubR1 ($BubR1^{H/H}$) show short lifespan, growth failure, sarcopenia, cataracts, fat loss and cardiac problems, and accumulate $p16^{Ink4a}$-positive senescent cells [7,8].
2. **BubR1 abundance can be raised pharmacologically.** BubR1 stability depends on NAD+ and the deacetylase SIRT2; the NAD+ precursor nicotinamide mononucleotide (NMN) increased BubR1 protein in vivo, and SIRT2 overexpression extended median lifespan of $BubR1^{H/H}$ mice [9].
3. **More BubR1 is protective.** Transgenic high BubR1 expression reduced aneuploidy and tumours and extended healthy lifespan in mice [10].

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3. Repositioning candidates (ranked)

Candidate A — NAD+ precursors: nicotinamide riboside (NR) or NMN

Best fit: BUB1B (MVA1) with an experimentally demonstrated unstable BubR1 protein that retains function when its abundance is restored. Evidence level: mechanism in mouse and cells; paediatric human safety data exist in another genome-instability disease.

- **Rationale:** NAD+ → SIRT2 → deacetylation of BubR1 (lysine 668) → more stable BubR1 [9]. Some MVA-associated missense BubR1 proteins are degraded unusually fast; others have functional defects despite normal stability [6]; for the unstable class, a drug that slows BubR1 turnover targets the specific defect. Restoring BubR1 levels rescues the checkpoint in patient cells [6].
- **Human experience:** NR has been studied in **children and adults with ataxia-telangiectasia** (A-T, another inherited genome-instability syndrome): in a Radboud UMC study of 24 patients (17 under 18), four months of weight-based oral NR was associated with improved ataxia scores and higher serum IgG [11]. The US FDA has since granted NR chloride **Orphan Drug and Rare Pediatric Disease designations for A-T** [12]. This gives a starting point for paediatric safety and dosing that MVA does not have.
- **Status:** NR and NMN are sold as dietary supplements in several countries; neither is an approved drug for any indication.
- **Mechanistic nuance (two acetylation sites):** BubR1 stability is controlled by acetylation at two different lysines with opposite effects. Deacetylation of K668 by SIRT2 protects BubR1 from age-related loss [9], whereas acetylation of K250 (mouse K243) protects it from APC/C-mediated degradation during mitosis, and mice that cannot acetylate this site have a weak checkpoint and more tumours [19]. Boosting NAD+/SIRT2 is therefore not simply “less acetylation = more BubR1”; the validation plan (Section 5) measures BubR1 levels directly rather than assuming the net effect.
- **Important caution:** plain **nicotinamide** (the vitamin B3 amide) is a sirtuin *inhibitor* in cells, and nicotinamide treatment *lowered* BubR1 protein in mouse fibroblasts [9]. It must **not** be used as a cheap substitute.
- **Not expected to help** if both BUB1B alleles are truncating (no protein to stabilise), or in CEP57 disease.

Candidate B — Intermittent senolytic: dasatinib (± quercetin)

Possible fit: BUB1B and TRIP13 degenerative/progeroid features. Evidence level: genetic proof-of-concept in BubR1 mice; drug efficacy shown only in other progeroid models.

- **Rationale:** genetic removal of $p16^{Ink4a}$-positive senescent cells in $BubR1^{H/H}$ mice delayed sarcopenia, cataracts and fat loss, and late-life clearance slowed already-established problems [7]. The drug combination dasatinib + quercetin kills several senescent cell types and improved healthspan in another progeroid mouse model ($Ercc1^{-/\Delta}$) [13].
- **Gentler alternative — fisetin:** among ten flavonoids screened, fisetin was the most potent senolytic; intermittent oral fisetin lowered senescence markers in several tissues of progeroid and aged mice, extended lifespan when started late in life, and reduced senescent cells in human fat-tissue explants [20]. It is a dietary supplement (not an approved drug, no paediatric trials) but has a milder expected toxicity profile than dasatinib, so it may be the better first molecule to test.
- **Status:** dasatinib is FDA-approved in children — for Ph+ chronic myeloid leukaemia (from 2017) and, with chemotherapy, for newly diagnosed Ph+ acute lymphoblastic leukaemia from age 1 (late 2018) [14] — so paediatric pharmacology and toxicity are well characterised. Quercetin is a supplement.
- **Cautions:** (i) genetic senescent-cell clearance did not extend lifespan in the cited BubR1 hypomorphic mouse experiment [7,9]; this does not establish whether senolytic drugs could affect survival; (ii) senescence is also a tumour-suppressive barrier and this child may have a high embryonal-tumour risk [4], so any senolytic study must include tumour surveillance; (iii) senolytic *drugs* have not yet been tested in a BubR1 model.
- **Overall:** moderate expected value, higher risk than Candidate A; should be tested in patient cells first.

Candidate C — Risk reduction (what to avoid or use carefully)

- **Minimise genotoxic exposure:** BubR1-insufficient mice are more sensitive to oxidative/carcinogen stress [15]. For tumour surveillance prefer **ultrasound and MRI** over CT where clinically equivalent.
- **Avoid HSP90 inhibitors (e.g. 17-AAG):** although they kill some aneuploid cancer cells, transient HSP90 inhibition *increased* chromosome instability and selected for new aneuploid karyotypes in yeast [21] — the wrong direction for a child whose checkpoint is already weak.
- **Growth hormone:** responses in MVA are inconsistent (poor overall effectiveness despite reported growth of 6–10 cm/year during treatment in one CEP57 patient, who also had high IGF-1, consistent with GH resistance [16]; early gain in another [17]); one report suggests intrinsic growth-plate resistance [18]. Given the cancer predisposition in BUB1B/TRIP13 disease, GH needs an explicit risk–benefit discussion and is not proposed here.
- **Tumour surveillance (not a drug, but highest impact):** for TRIP13 and BUB1B genotypes, embryonal tumour risk is high [4]; a structured paediatric surveillance protocol following the SIOP-Europe consensus recommendations for Wilms tumour surveillance in at-risk children [22] (renal ultrasound, which is also non-ionising) is likely the single most life-protecting measure.

Other candidates considered (lower priority)

Compound	Why considered	Why lower priority
Metformin (approved, paediatric use)	AMPK activation; reduced oxidative damage selectively in trisomy-21 fibroblasts [23]	Cell data only, no MVA data; weight/growth effects in an already small child
Chloroquine (approved)	Selectively slowed proliferation of aneuploid cells in a screen [24]	Aneuploid-cell *killing* strategy, relevant only if a tumour arises; retinal/cardiac toxicity
Rapamycin	Plausible ageing/mTOR link	Baker et al. reported preliminary unpublished data of similar progression of ageing phenotypes in progeroid BubR1 mice treated with rapamycin vs. controls [25]; not proposed
Spermidine, HDAC inhibitors, antioxidants	Plausible ageing/autophagy links	No source linking them to BubR1, TRIP13 or CEP57 models; not proposed

4. Genotype decision table (plugs directly into a Track 1 answer)

Confirmed causal variant	First candidate	Second	Key caveat
BUB1B, unstable BubR1 with demonstrable residual function	**A** (NR/NMN)	B (cells first)	Avoid high-dose nicotinamide
BUB1B, both alleles truncating	B (cells first)	—	A has no protein to stabilise
TRIP13 (loss of function)	B (cells first)	—	Wilms tumour surveillance is the priority
CEP57	None strongly supported	—	Minimal SAC defect; focus on supportive care

The Track 1 output fields (gene, `p_id`) give a first pointer: a `p.` change such as `p.Xxx123Yyy` indicates missense and suggests the first row, but whether the protein is actually unstable (and still functional when restored) must be measured in the patient's cells (Section 5, step 1) before choosing Candidate A; `*`, `fs` or `Ter` indicates truncating.

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5. Proposed validation plan (cheap, fast, reusable)

All steps use patient-derived cells obtainable from a skin biopsy or blood (fibroblasts or lymphoblastoid lines), with parental cells as controls.

1. **Target engagement (Candidate A):** treat cells with NR or NMN; measure NAD+ and BubR1 protein (western blot) and BubR1 half-life (cycloheximide chase). Include nicotinamide as a negative control (expected to lower BubR1 [9]).
2. **Functional rescue:** SAC strength (time in mitosis under nocodazole, live imaging) and aneuploidy rate by FISH or single-cell sequencing, before vs. after treatment.
3. **Senescence burden (Candidate B):** SA-β-gal and p16 staining in serially passaged fibroblasts; test dasatinib ± quercetin for selective killing of senescent versus proliferating cells.
4. **Go/no-go:** only candidates that raise BubR1 and lower aneuploidy *in the child's own cells* move to clinical discussion.

Scalability: the decision table and assay pipeline are gene-agnostic for SAC disorders and can be reused for any future MVA family, or for other conditions where a missense protein is unstable. The decision table is implemented as a small open command-line tool (decide.py) that takes a gene and HGVS p. variants, or a Track 1 predictions file, and returns the candidate branch; a companion script (check_refs.py) re-verifies that every DOI in this report is registered. Code, pitch video and methods form: https://github.com/ivanfrancesco2026-cpu/mva-hackathon-2026-track2

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6. Limitations

- No access to, or use of, the child's genome or clinical data; recommendations are conditional on genotype.
- Evidence for both main candidates comes from mouse models, cell biology and a different human disease (A-T), not from humans with MVA.
- Rapid literature review (not systematic); relevant studies may be missing.
- Prepared by a second-semester electromechanical engineering student with AI assistance; requires expert verification.

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